# Bank Marketing Data Set

*Data Visualization & Predictive Modeling for Customer Conversion in Bank Marketing Campaigns*

**Sector : Fintech**

Github Tableau

**G-11**

DATA CLEANING

DATA TRANSFORMATION

DATA ANALYSIS


**Team Members:**

Jagruthi Pulumati - 2401010196
Abhyanth K - 2401010012
Kasula Lalithendra - 2401010216
Sachin Jaiswal - 2401010399
Ankit Raj Singh - 2401010075
Varun Sharma - 2401010497

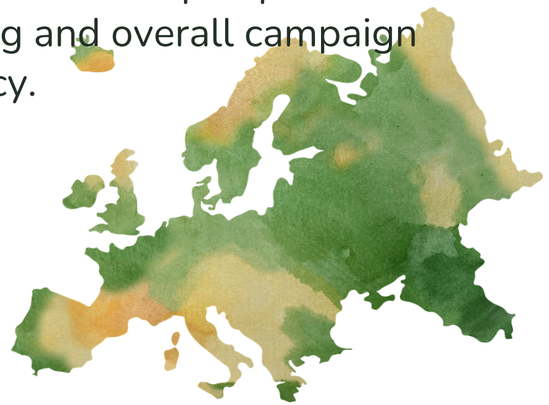
**Faculty: Archit Raj**

# Bank Marketing Campaign Analytics

Optimising term-deposit conversion through customer, campaign & macro-economic analytics

## Context

- European banks use term deposits as a key and stable source of funding.
- Marketing campaigns, especially through phone calls, are widely used to reach potential customers.
- However, these campaigns involve high operational costs and typically result in low conversion rates.
- Understanding customer behavior and identifying patterns in responses can help improve targeting and overall campaign efficiency.



## Problem Statement

- The current campaign conversion rate is low, meaning a large number of customer contacts do not result in successful subscriptions.
- This leads to inefficient use of resources and increased operational costs.
- There is a need to analyze the data to identify which customer and campaign factors influence subscription decisions, so efforts can be focused on high-potential customers.

**Avg Call Duration**

**Conversion Rate**

**Avg Contact per Customer**

## Objective

- To analyze the bank marketing dataset using data cleaning, transformation, and visualization techniques.
- To identify key customer and campaign features that impact term deposit subscription.
- To derive insights that can help improve targeting strategies and increase campaign effectiveness.

# DATA ENGINEERING

## DATASET OVERVIEW

**41188 X 21**

Total Records

**41176 X 27**

Cleaned Dataset

## DATA CLEANING PROCESS

- 12 exact duplicate rows were identified and removed, reducing the record count from 41,188 to 41,176 unique rows.
- Cryptic column names were renamed for clarity (e.g., poutcome to previous_campaign_outcome).

## KEY COLUMNS

[subscription_flag] [contact_recency] [prev_success_flag] [previous_campaign_outcome] [call_duration_sec] [contact_type]

### RAW DATA

**41,188 records × 21 columns UCI Bank Marketing Dataset - Moro, Cortez & Rita (2014)**

↓

### FILTERING AND VALIDATION

**Filtering involved removing 12 duplicate records, capping call-intensity outliers at 10, and merging sparse categories like "illiterate" to reduce analytical noise.**

↓

### DATA CLEANING

**Removed 12 duplicate records, standardizing "Unknown" values, capping call-count outliers at 10, and merging sparse education categories to ensure data integrity.**

↓

### FEATURE ENGINEERING

**Added age_group (5 buckets) · call_intensity · contact_recency · rate_environment · prev_success_flag · education_ordinal · region_NUTS_II.**

↓

### TRANSFORMATION AND STRUCTURING

**Standardized variables and engineered age, rate, and education segments to optimize analytical interpretability.**

### Final Dataset

**41176 clean records × 27 features ready for analysis & visualization.**

# KPI & Metric Framework

## KPIs Measured

**41,176**
Total Customers

**11.27%**
Success Rate

**4,639**
Successful Customers

**4.3**
Avg Call Duraction

**14.74%**
Cellular Conversion Rate

**65.11%**
Prior Success Conversion Rate

## Why These Metrics?

These KPIs directly measure the ultimate success of the marketing campaign (Conversion Rate) alongside the operational efficiency of the sales team (Call Intensity and Duration).

### Mapped KPIs

- Conversion Rate & Prior Success Lift
- Call Intensity, Average Call Duration.
- Previously Contacted,,Cold-Contact Conversion
- Comparative Conversion Yield across Demographic, Occupational, and Educational Segments.

# Exploratory Data Analysis Insights

**Campaign Performance:** Conversion rate is low (~11.27%), indicating most customer contacts do not result in subscription. This highlights inefficiency in current targeting strategies.

**Customer Interaction Impact:** Customers with previous successful outcomes show ~65% conversion, while cold contacts are around ~8.83%.
Past interaction history is a strong predictor of future conversion.

**Contact Strategy:** Cellular communication (~14.74%) outperforms telephone (~5.23%) in conversion.
Also, longer call duration is associated with higher success rates.

**Timing & Seasonality:** Conversion varies significantly across months, with certain periods showing higher performance.
Campaign timing plays an important role in effectiveness.

**Economic Influence:** Conversion is higher in low interest rate environments (~24%) and drops sharply in high-rate conditions (~4.83%).
Macroeconomic factors impact customer decision-making.

**Customer Segmentation**: Major customer groups include admin, blue-collar, and technician roles.
However, demographics alone are weaker predictors compared to behavioral factors.

**Overall Insight:**
Customer engagement history, campaign strategy (channel & duration), and economic conditions are the key drivers of conversion, while demographic features have a comparatively lower impact.

# Advanced Analysis

## Segmentation Analysis – Customer Conversion Segments

*Customers are segmented based on engagement history, campaign interaction, and response behavior.*

| Segment | Criteria | % of Population | Key Characteristcs |
|---|---|---|---|
| High Potential | Previous success, recent contact, higher duration | ~10–15% | Highest conversion rates (~65%), strong campaign response |
| Moderate Potential | Multiple contacts, no prior success, moderate engagement | ~30–40% | Moderate conversion (~10–12%), with potential improvement through better targeting and contact strategy |
| Low Potential | Cold contacts, no history, low duration | ~50–60% | Lowest conversion (~8–9%), high outreach cost, low ROI |

### Scenario Analysis

If campaign targeting is optimized:

- Conversion rate can increase from ~11.27% → ~15–18%
- Reduction in unnecessary contacts by 20–30%
- Higher ROI through focused outreach on high-potential segments

**Insight:**

Targeted campaign strategy can significantly improve conversion efficiency and reduce wasted effort.

### Key Insights

- Customers with prior successful interactions (~65%) are the strongest predictors of conversion

(Analyzed using: Pandas, Seaborn — groupby analysis & visualization)

- Contact strategy (channel, duration, frequency) plays a critical role beyond demographics

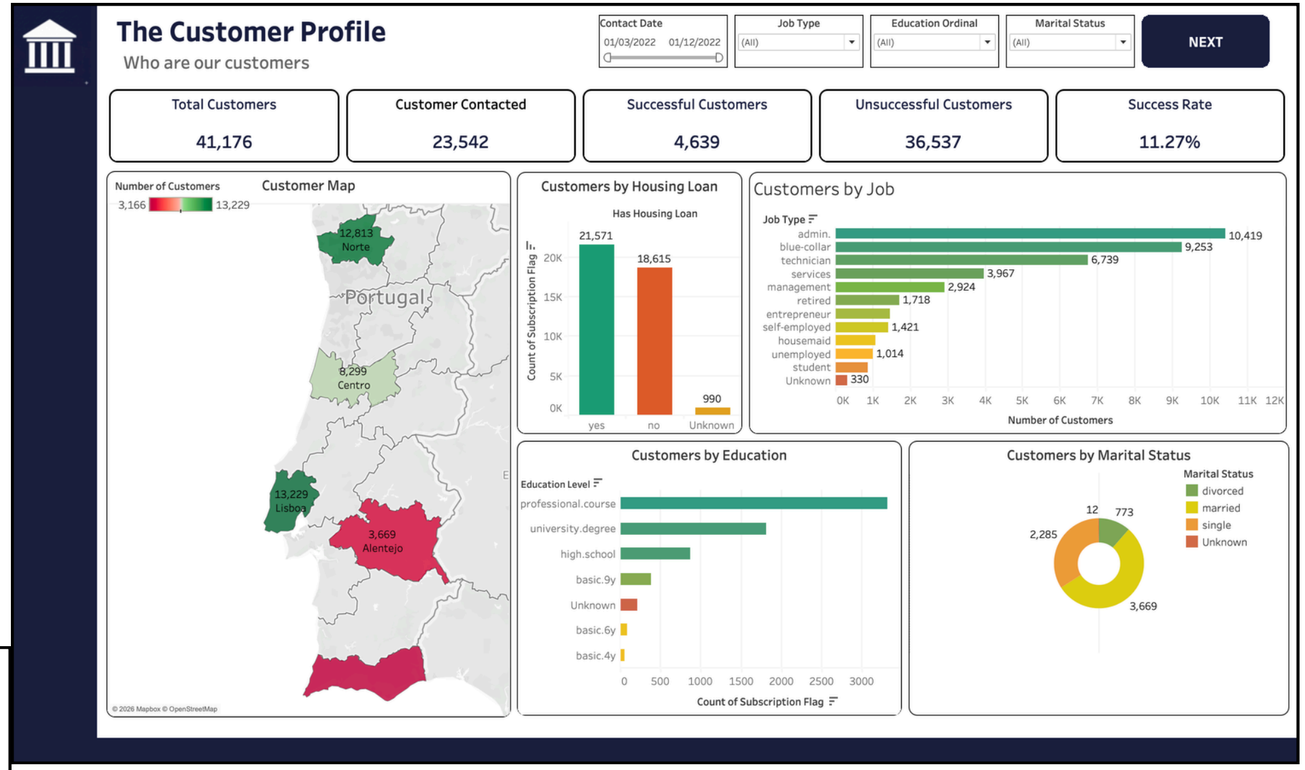
(Analyzed using: Pandas, Matplotlib — feature comparison & distribution analysis)

- Economic and timing factors influence performance, but engagement history is the most impactful driver

(Analyzed using: Pandas, NumPy — trend analysis & aggregation)

# 👤 Customer Profile

- Majority of customers are concentrated in admin, blue-collar, and technician job segments
- Total base includes 41,176 customers with 4,639 conversions
- Customer distribution differs across regions (e.g., Lisboa, Norte, Alentejo)

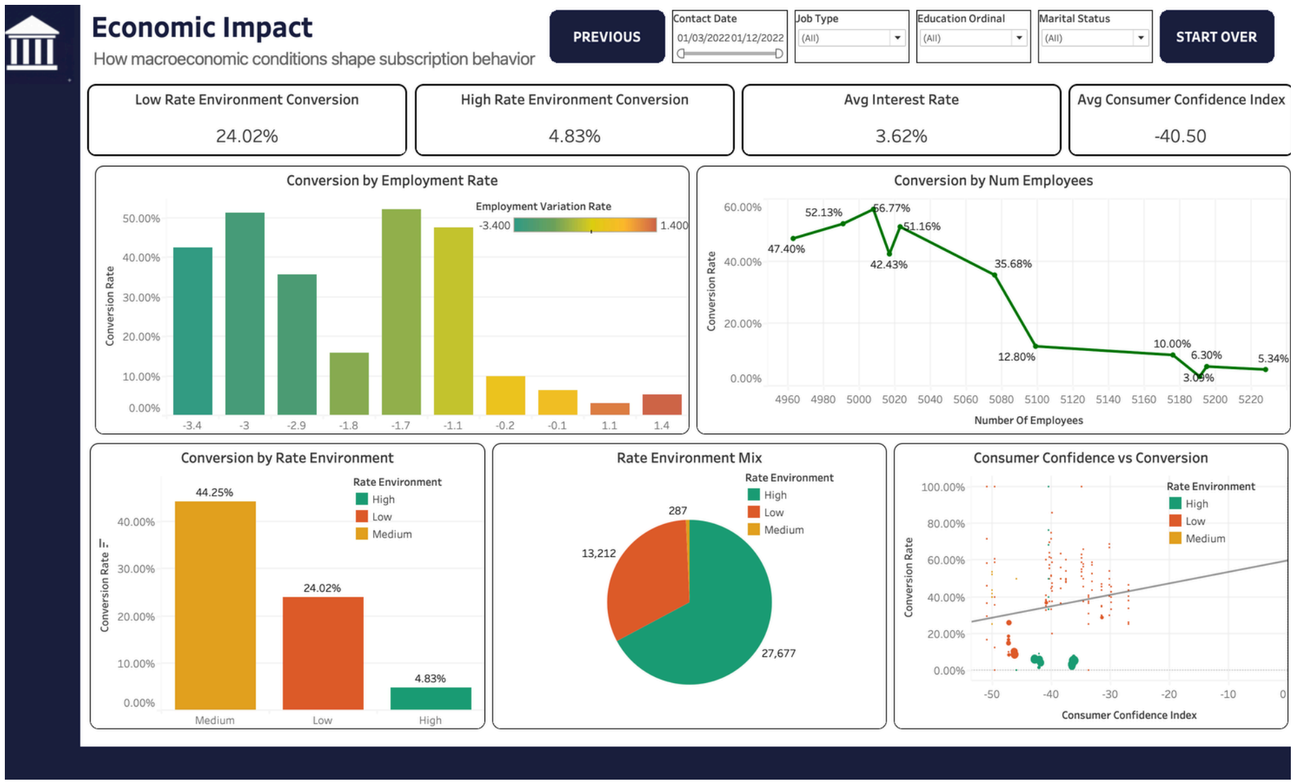


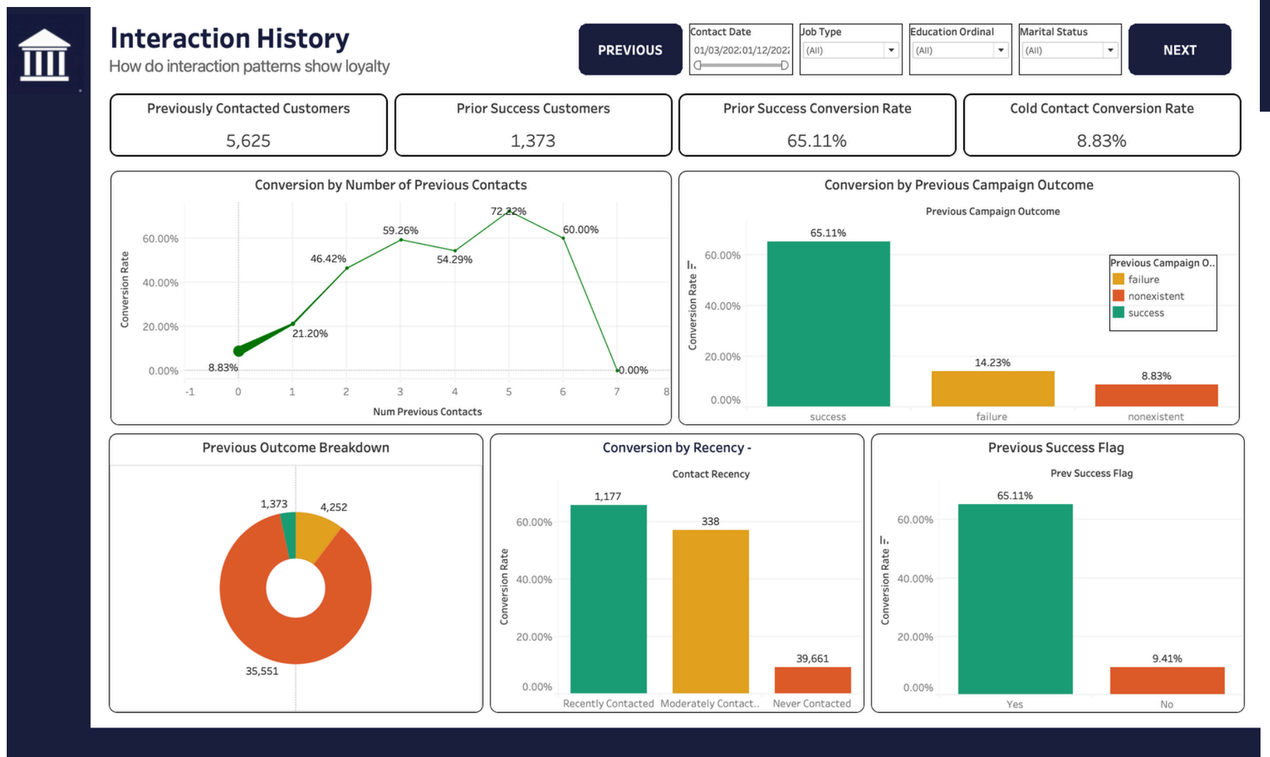
# 📣 Campaign Effectiveness

- Overall conversion remains low (~11.27%), indicating uneven campaign performance
- Cellular channel (14.74%) outperforms telephone (5.23%) significantly
- Conversion varies across months, showing timing differences in campaign success

# Economic Impact

- Conversion is significantly higher in low interest rate conditions (24.02%) compared to high-rate environments (4.83%)
- Economic indicators like employment variation and consumer confidence show noticeable variation in conversion levels
- External economic conditions clearly influence customer decision behavior




# Interaction History

- Customers with previous successful outcomes have the highest conversion (~65.11%)
- Conversion improves with more interactions, peaking around 5 contacts (~72%)
- Cold contacts (~8.83%) have the lowest conversion, indicating weaker engagement

# Recommendations

1

**Prioritize High-Probability Customers**

Customers with previous successful interactions show ~65% conversion, much higher than cold leads.
Focus outreach on previously engaged customers to improve campaign efficiency.

2

**Optimize Contact Strategy**

Conversion improves with repeated contacts but peaks around 5 interactions (~72%) before dropping.
Limit excessive follow-ups and optimize the number of contacts per customer.

3

**Shift to High-Performing Channel**

Cellular communication shows ~14.7% conversion vs ~5.2% for telephone.
Allocate more resources to cellular campaigns for better outcomes.

4

**Align Campaigns with Timing**

Conversion rates vary significantly across months and economic conditions.
Schedule campaigns during high-performing periods and favorable rate environments.

5

**Improve Overall Conversion Efficiency (High Impact)**

With an overall conversion rate of ~11.27%, even small improvements can drive significant gains.

- Target high-response segments to increase conversions
- Reduce low-performing outreach to save costs
- Improve call quality (duration + engagement)

# Impact & Value

### Campaign Cost Optimization

- ~89% of calls do not convert (11.27% baseline), indicating high wasted outreach cost
- Targeting high-probability customers (e.g., prior success ~65%) improves ROI per contact
- Shifting to cellular (14.74% vs 5.23%) reduces cost per successful conversion

### Operational Efficiency

- Focusing on previously engaged customers increases conversion from ~8.83% to ~65.11%
- Optimizing contact frequency (peak ~72% at 5 contacts) prevents over-calling inefficiency
- Data-driven filtering reduces unnecessary outreach and improves agent productivity

### Revenue Growth Potential

- Improving conversion beyond 11.27% baseline directly increases subscription revenue
- Leveraging high-performing segments and channels boosts overall campaign yield
- Better targeting converts existing customer base more effectively without increasing volume

### Strategic Decision-Making

- Economic conditions (low-rate ~24% vs high-rate ~4.83%) guide campaign timing strategy
- Customer and campaign insights enable more precise segmentation and targeting
- Dashboard-driven insights support faster, data-backed marketing decisions

# Why Approve This?

**Improves conversion efficiency, reduces campaign costs, and enables smarter, data-driven marketing decisions.**

# Limitations And Future Scope

## Data Issues

**Missing & Unknown Values**

- Several fields (job, education, housing, loan, poutcome) contain “unknown” categories
- Handling these during cleaning may affect model accuracy and insights

**Imbalanced Target Variable**

- Dataset is skewed with low conversion rate (~11%) vs majority non-conversions
- This imbalance can bias analysis toward the majority class

**Limited Customer Context**

- Dataset lacks deeper features like income, digital behavior, or past financial activity
- Limits understanding of true customer intent and segmentation

## Future Data Enhancements

**Predictive Modeling**

- Build classification models (Logistic Regression, Random Forest) to predict high-probability customers
- Improve targeting before campaign execution

**Real-Time Campaign Dashboard**

- Extend current dashboard to real-time monitoring of campaign performance
- Enable quick decision-making and campaign adjustments

**Enhanced Data Integration**

- Incorporate additional data like customer transaction history, income level, or behavioral data
- Improve segmentation and prediction accuracy